# Programming Fundamentals II: Lab 09

**Due Date: Tuesday, November 12 at Midnight**

---

**Part 1. Train System (10 Marks)**

Suppose we have a system designed for managing travel via rail. In our system we have Trains, Persons and Cargo. Both persons and cargo are able to be transported on a train. We want a train to be able to track everything on it in a single data structure.

Persons will track their name and age (int). Cargo will track its id (String) and weight (in kilograms).

Persons and Cargo have little in common, so it doesn't fit to use a common parent class for the two. Here will use an interface call Transportable to enforce the Person and Cargo classes to implement any methods necessary to board a train. Everything that is being transported by a train must implement a calculateFare() method. They must also implement an identify() method that will return a name for a person, or an id for cargo.

For a Person, fare is determined by the Person's age. Anyone 18 or under is $15.00, anyone 65 or older is $10.00, and everyone else is $25.50. The rate for transporting cargo is $1.50 per kilogram.

You do not need to code any getters/setters that will not be used for this assignment. You may need to add fields that are not present in the description. Use the included Driver.java to test your code.

a. **(2 Marks)** Write the code for the Transportable interface.

b. **(4 Marks)** Write the code for the Person and Cargo classes. The classes must implement the Transportable interface. Make sure you implement any methods and fields that the interface needs. Include a no-arg constructor, and the constructor required by Driver.java

c. **(4 Marks)** Write the code for the Train class. The Train class should have name and an ArrayList that stores the objects being transported by the Train. The addTransport() method will add a **either a person or cargo** to the list. The displayTransport() method will print out everything on the train and their transportation costs.

Add the **date and your name and student number** to the top of your java files. Please submit your java files to Nexus before the end of the day on the lab date. **Late submissions will be not be accepted.**